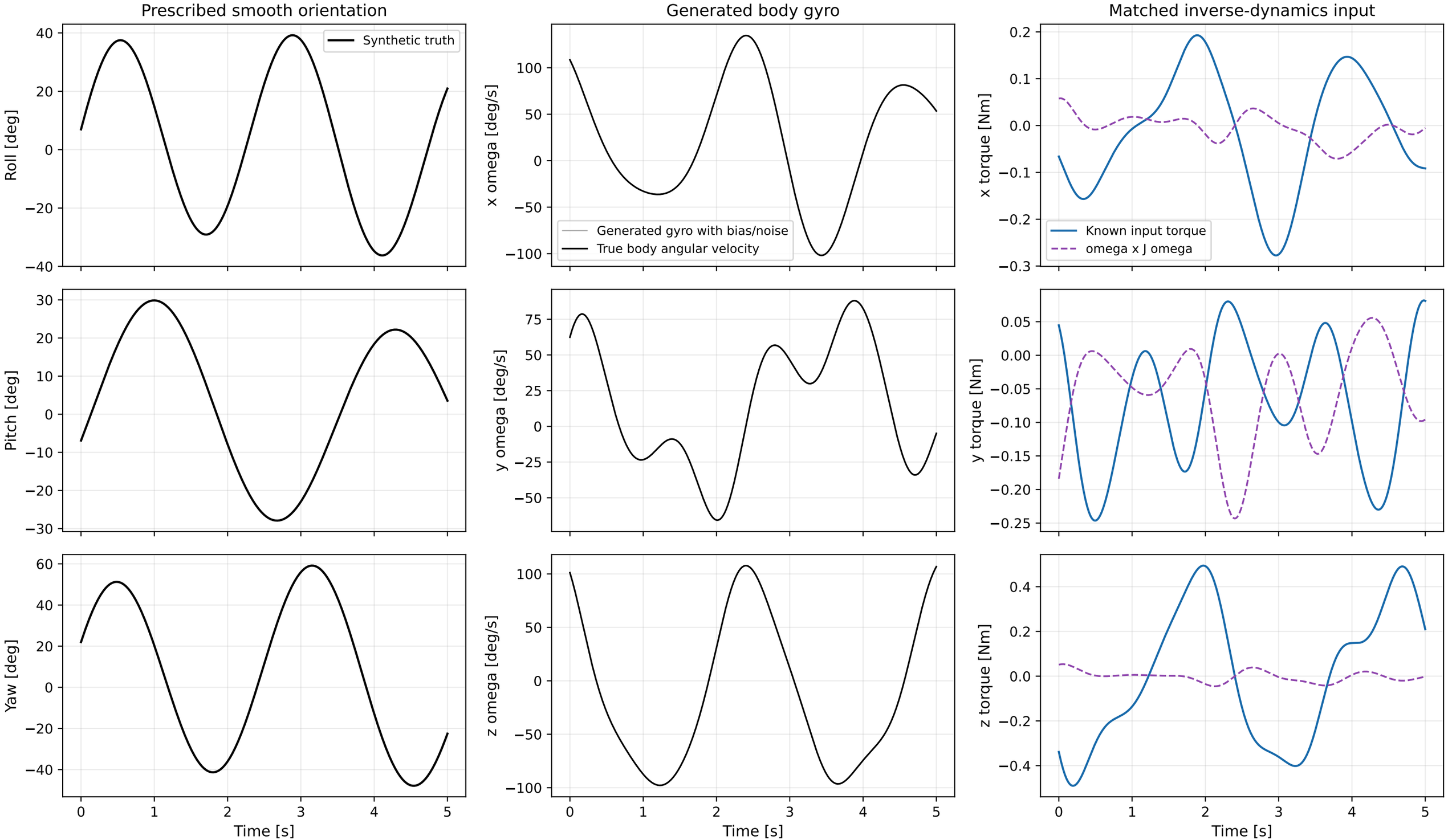Synthetic rotation inputs: same inertia and known torque for both methods
Prescribed kinematic motion; not a six-DoF flight simulation